

Awantika AIparc 4

March 30, 2026

```
[9]: def is_safe(board, row, col, n):
    for i in range(row):
        if board[i] == col or abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve_nqueens(n):
    solutions = []
    board = [-1] * n

    def backtrack(row):
        if row == n:
            solutions.append(board.copy())
            return

        for col in range(n):
            if is_safe(board, row, col, n):
                board[row] = col
                backtrack(row + 1)
                board[row] = -1

    backtrack(0)
    return solutions

n = int(input("Enter value of N: "))

solutions = solve_nqueens(n)

print("\nTotal solutions:", len(solutions))

for sol in solutions:
    print("\nSolution:")
```

```
for i in range(n):
    for j in range(n):
        if sol[i] == j:
            print("Q", end=" ")
        else:
            print("O", end=" ")
    print()
```

Enter value of N: 4

Total solutions: 2

Solution:

O Q O O
O O O Q
Q O O O
O O Q O

Solution:

O O Q O
Q O O O
O O O Q
O Q O O